2026-06-10-SP1-model-access-plan

SP1 — Model-Access Refinement: Implementation Plan

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Maintainer	SP1 planning subagent (READ-ONLY planning pass)
Source spec	`docs/superpowers/specs/2026-06-10-llms-access-master-roadmap.md` §SP1
Evidence base	`docs/superpowers/specs/analysis/2026-06-10-B-model-access.md`
Authority	Cascades from `constitution/` + root `CLAUDE.md/CONSTITUTION.md`; anti-bluff §11.4 governs every closure

> **Anti-bluff note (§11.4 / §11.4.123):** This is a *plan*, not a claim of done work. No task below is "complete" until it carries captured runtime evidence per §11.4.5 / §11.4.69 / §11.4.107. Brainstorming HARD-GATE: no implementation, no push until this plan is operator-approved. Subagent-driven execution is the default (§11.4.70). TDD RED-first discipline (§11.4.43 / §11.4.115); every closed defect gets a standing regression guard (§11.4.135).

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1. Goal & scope

Goal (verbatim from request): *"For every provider, if an API key is provided OR recognized from* `.env` */* `.api_keys` */ shell-exported, that provider's WORKING models become available — everything dynamically obtained + validated + verified via LLMsVerifier."*

SP1 closes five evidence-backed defects (roadmap §1a): - **D-2** CLI `handleListModels` lists `failed/pending` models as available (`cmd/cli/main.go:1361/:1392`). - **D-3** `secrets.LoadAPIKeys` is DEAD code, never called in prod (`internal/secrets/loader.go:30`). - **D-4** working-model filter (`Verified ∧ score≥min`) loaded but never applied (`adapter.go:175`). - **D-5** hardcoded provider model lists (CONST-036) (`openai_provider.go:203`, `anthropic_provider.go:205`, …). - Plus the **no-key-presence-gate** gap (a user with one key sees every provider) and the **single-provider factory** gap (`factory.go:9` builds ONE provider by `config.Type`, no key-present enumerator).

Out of scope (owned by other SPs): HelixAgent `/v1/completion/models` hardcoded list (D-1 → SP2); CLI-agent stub packages (D-6 → SP4); `go.mod replace dev.helix.agent` wiring (D-7 → SP4/SP5); constitution Rev-table drift (D-8 → SP0).

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2. Confirmed evidence anchors

All anchors below were opened and confirmed in this planning pass (file:line as read).

Anchor	Confirmed fact
`helix_code/internal/secrets/loader.go:30`	`func LoadAPIKeys() error` reads `$HOM` `api_keys.sh` (L33-37 `loadFromShell`), e `.env` (L119 `findEnvFile`, L84 `loadFromE` `os.Setenv` each (L78/L103). CONST-042-cle logging). Zero prod call-sites = DEAD.
`helix_code/internal/config/config.go:307-345`	`v.AutomaticEnv()` + `SetEnvPrefix("H` (L308-309); 8 single-alias per-provider bindings openai/anthropic/gemini/deepseek/groq/mistral/x narrower than helix_agent's multi-alias.
`helix_code/cmd/cli/main.go:1355-1390`	`handleListModels`: P1 verifier `GetVerif` (L1361) → `printVerifiedModels` (prints single `c.llmProvider.GetModels()` (L1 `verifier.FallbackModels` (L1387).
`helix_code/cmd/cli/main.go:1392-1414`	`printVerifiedModels` renders `✓ verif` `pending` / `× failed` / `⏳ rate-limited` failed/pending rows as "available" (D-2).
`helix_code/internal/verifier/adapter.go:175-180`	`GetMinAcceptableScore()` default 6.0 fr `Scoring.MinAcceptableScore`. **Never i** listing path.
`helix_code/internal/verifier/adapter.go:183-224`	`GetVerifiedModels`: gated on `IsEnable` cache→live→stale→`getFallbackModels(` `filterByProviderConfig(models)`. No Verified/score filter.

Anchor	Confirmed fact
helix_code/internal/verifier/adapter.go:277-289	filterByProviderConfig drops only config.Providers[m.Provider].Enal (manual flag).
helix_code/internal/verifier/types.go:27/31/32/33/48/54	VerifiedModel{DisplayName, Verif VerificationStatus string, Conte OverallScore float64, LastVerifi "working" fields.
helix_code/internal/llm/openai_provider.go:201-239	initializeModels() hardcodes 4 models (4/3.5-turbo). CONST-036 risk.
helix_code/internal/llm/anthropic_provider.go:203-205+	getAnthropicModels() hardcodes claude- families. CONST-036 risk.
helix_code/internal/llm/factory.go:9-101	NewProvider(config) switch builds ONE config.Type. InitializeModelManage configs but only those with config.Enabl every key-present provider" enumerator.
submodules/helix_agent/internal/verifier/provider_types.go:348-385+	SupportedProviders map[string]*ProviderTypeInfo with p alias EnvVars (claude {ANTHROPIC_API_KEY,CLAUDE_API_KE {GEMINI_API_KEY,GOOGLE_API_KEY,A …). THE reuse template.
submodules/helix_agent/internal/verifier/startup.go:389-414	Per-provider loop: for each EnvVar, os.Gete && !isPlaceholder → DiscoverModel providerType, apiKey) dynamic-first, st info.Models fallback. THE key→discover fl
submodules/llms_verifier/llm-verifier/api_keys/{env_scanner.go,manager.go,priority.go}	EnvVarScanner + faulty-key registry. Import as digital.vasic.llmsverifier/api_

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3. Target design

The working-model funnel (analysis §4.1, anti-bluff-tightened):

```
startup (cmd/server/main.go AND cmd/cli/main.go) — BEFORE config load
  └─ secrets.LoadAPIKeys()                          [D-3 — WIRE; loader.go:
        reads $HOME/api_keys.sh → else walked-up .env → os.Setenv each
        (precedence note: already-exported shell env is NOT overwritten by
  └─ config.Load()  (viper AutomaticEnv + BindEnv — now sees loader-popula
  └─ keyrec.PresentProviders()                      [NEW — internal/llm/key
        for each provider, any(os.Getenv(alias)) != "" && !isPlaceholder(v
        alias table sourced from helix_agent SupportedProviders.EnvVars (§
        → set[providerType]bool
  └─ verifier.Adapter.GetWorkingModels(ctx, present) [NEW METHOD — adapter
        m kept  iff  present[m.Provider]                        (key-pre
                AND  m.Verified == true                         (types.g
                AND  m.VerificationStatus == "verified"         (types.g
                AND  m.OverallScore >= GetMinAcceptableScore()    (adapter
```

```
              AND  config.Providers[m.Provider].Enabled != false (existi
  └─ expose:
        - CLI handleListModels (main.go:1361) → GetWorkingModels; printVer
        - server handlers (internal/server/handlers.go:1049/1140/1206)
        - model_manager registry (model_manager.go:132 SetVerifierAdapter
  └─ verifier-DISABLED / cold path  [DECISION-2]
        NOT FallbackModels-as-working (that = §11.4 / CONST-035 PASS-bluff
```

De-hardcoding (D-5, CONST-036): convert each hardcoded `initializeModels()` to either (a) live catalog fetch (mirror OpenRouter `fetchCatalog` / Ollama `/api/tags`) OR (b) defer to the verifier catalog. The working-model funnel above already routes the *displayed* list through the verifier, so the provider `GetModels()` lists stop being the user-facing source — but they remain a CONST-036 violation as long as they ship literal slices and are reachable via CLI P2 fallback (main.go:1372). See DECISION-3.

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4. Decisions needing operator input

These are genuine ambiguities the agent must NOT guess (§11.4.6 / §11.4.101 — surface via `AskUserQuestion` per §11.4.66 before the affected task starts):

- **DECISION-1 — LoadAPIKeys precedence vs shell-exported env.** `loader.go` calls `os.Setenv` unconditionally, so `.env/api_keys.sh` values would OVERRIDE an already-exported shell var. The request says "provided OR recognized from `.env` / `.api_keys` / shell-exported" without ranking them. Options: (a) shell-export wins (loader only fills GAPS — safest, least surprising); (b) file wins (loader overrides — current loader behaviour); (c) explicit precedence `.env` < `api_keys.sh` < shell. **Recommended: (a) gap-fill** — least destructive, matches "recognize if not already set". Affects T1.1.1 + a loader change.

- **DECISION-2 — verifier-disabled behaviour** (`HELIX_VERIFIER_ENABLED=false` default, `.env.example:30`). When the verifier is off there is NO authoritative "working" signal. Options: (a) "working models" require the verifier — disabled ⇒ empty list + honest notice ("enable HELIX_VERIFIER_ENABLED to see working models"); (b) fall back to per-provider LIVE `GetModels()` for key-present providers only (never hardcoded fallback), labelled `unverified`; (c) keep today's `FallbackModels` path. (c) is a §11.4/CONST-035 PASS-bluff and is REJECTED. **Recommended: (a)** for the headline "working models" command + (b) behind an explicit `--include-unverified` flag. Affects T1.3.3.

- **DECISION-3 — de-hardcoding strategy per provider** (CONST-036). Options: (a) delete the literal `initializeModels` slices and have `GetModels()` fetch live (OpenAI `/v1/models`, Anthropic `/v1/models`, etc.) — most faithful to CONST-036 but adds network dependency + needs the API key at construction; (b) keep a thin static slice ONLY as an offline fallback but route ALL user-facing listing through the verifier funnel (§3) so the literal is never shown as authoritative; (c) full per-provider live fetch now for the 8 cloud providers, defer Azure/Bedrock/VertexAI/Copilot/Qwen. **Recommended: (b)+(c) staged** — funnel first (removes the user-facing bluff cheaply), then convert OpenAI/Anthropic/DeepSeek/Mistral to live-fetch. Note §11.4.124: the hardcoded slices are NOT dead code (CLI P2 reaches them) — investigate-before-remove is satisfied; we convert, not silently delete.

- **DECISION-4 — key-recognition table source.** Options: (a) `import` helix_agent `SupportedProviders` directly (needs `replace dev.helix.agent` — D-7, owned by SP4/SP5, not yet wired); (b) lift a decoupled copy of just the {provider → []envVarAlias} map into a new HelixCode internal/llm/keyrecognition.go now, with a forward-note to converge on the shared substrate when D-7 lands. **Recommended: (b)** — unblocks SP1 without taking the SP4/SP5 dependency; the map is small, data-only, and CONST-051-decoupled. Record Catalogue-Check: extend helix_agent@<sha> per §11.4.74.

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5. Files to create / modify

Path	Action	Purpose
helix_code/internal/llm/keyrecognition.go	**create**	PresentProviders() + multi-alias {provider→[]envVar} table + isPlaceholder(); lifted/decoupled from helix_agent SupportedProviders.EnvVars (§11.4.74, DECISION-4).
helix_code/internal/llm/keyrecognition_test.go	**create**	unit: alias precedence, absence, placeholde rejection (mocks OK — unit only).
helix_code/cmd/server/main.go	**modify**	call secrets.LoadAPIKeys() first ir main() (D-3, T1.1.1).
helix_code/cmd/cli/main.go	**modify**	call secrets.LoadAPIKeys() first ir main() (D-3); handleListModels:1361 → GetWorkingModels; printVerifiedModels:1392 stops emitting failed/pending as available (D-2); P3 fallback gated by DECISION-2.
helix_code/internal/secrets/loader.go	**modify (DECISION-1 only)**	optional gap-fill precedence (skip os.Setenv when var already set).
helix_code/internal/verifier/adapter.go	**modify**	new GetWorkingModels(ctx, present map[string]bool) ([]*VerifiedModel, error) applying key∧Verified∧status∧min-score (D-4, uses GetMinAcceptableScore:175); kee GetVerifiedModels for raw catalog.
helix_code/internal/verifier/adapter_test.go	**modify/create**	unit: filter predicate truth-table + paired §1 mutation (strip key-gate → no-key provide leaks → FAIL).
helix_code/internal/llm/factory.go	**modify**	add BuildPresentProviders(configs present) enumerator (build every key-present provider, not ONE by config.Type).
helix_code/internal/llm/openai_provider.go	**modify (D-5/ DECISION-3)**	convert initializeModels:201 to live-fetch or verifier-deferred.

Path	Action	Purpose
helix_code/internal/llm/anthropic_provider.go	**modify (D-5/ DECISION-3)**	convert getAnthropicModels:203 likewise.
helix_code/internal/llm/{deepseek,mistral}_provider.go	**modify (D-5)**	convert hardcoded lists.
helix_code/internal/server/handlers.go	**modify**	:1049/:1140/:1206 route through GetWorkingModels + present-set.
helix_code/internal/llm/model_manager.go	**modify**	:132 SetVerifierAdapter/:202 getAvailableModels populate regist from working-model funnel, not hardcoded GetModels().
helix_code/tests/integration/sp1_model_access_test.go	**create**	-tags=integration, REAL verifier + REAL provider key, NO mocks (CONST-050).
helix_code/tests/e2e/challenges/sp1_working_models/...	**create**	one-key→only-that-provider→real-genera E2E (§11.4.107 liveness).
helix_code/tests/stresschaos/sp1_model_access_stress_test.go	**create**	verifier-unreachable / concurrent-list (§11.4.85).
helix_code/tests/regression/sp1_working_model_guard_test.go	**create**	standing RED_MODE guards for D-2/D-3 D-4 (§11.4.135).
challenges/banks/sp1_model_access/...	**create**	cross-cutting Challenge bank entry (§11.4.83 transcript).
helix_code/docs/guides/model-access.md (+ .html/.pdf)	**create**	user manual (CONST-066 always-sync).
.env.example	**modify**	document api_keys.sh + .env auto-load + per-provider aliases + verifier-disabled behaviour.
docs/ARCHITECTURE.md	**modify**	working-model funnel flow.
docs/Issues.md / docs/Issues_Summary.md / docs/CONTINUATION.md	**modify**	ATM-NNN entries per §11.4.54 / CONST-044.

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6. Phased TDD task list

Each task: **(a) RED** (reproduce gap on current code, RED_MODE=1) → **(b) IMPL** (exact file/func) → **(c) GREEN+evidence** (RED_MODE=0 guard + captured runtime proof) → **(d) rollback**.

Phase P1.0 — Pre-work (blocking)

- **T1.0.1** Resolve DECISION-1..4 via AskUserQuestion (§11.4.66). Capture answers into this plan (bump Rev).
- **T1.0.2** Catalogue-Check (§11.4.74): record extend helix_agent@<sha> + reuse llms_verifier/api_keys@<sha> in the ATM-NNN tracker entry.

- **T1.0.3** `git fetch --all --prune` + `git log HEAD..@{u}` (§11.4.37) before any edit; capture output.

Phase P1.1 — Key auto-recognition (D-3)

- **T1.1.1 — Wire `secrets.LoadAPIKeys` at startup.**
  - 1. RED: `tests/integration/sp1_model_access_test.go::TestLoadAPIKeys_WiredAtStartup` — set ONLY a key in a temp `.env` (no shell export), boot CLI/server entrypoint, assert the provider becomes key-present. `RED_MODE=1` on current tree FAILS (loader never called → key not recognized).
  - 1. IMPL: call `secrets.LoadAPIKeys()` as the first statement of `main()` in `cmd/server/main.go` and `cmd/cli/main.go`, BEFORE `config.Load()`. Apply DECISION-1 precedence in `loader.go` if chosen.
  - 1. GREEN: `RED_MODE=0` guard passes; capture stdout proving the temp-`.env` key was recognized (value never printed — CONST-042). Evidence path `docs/qa/<run-id>/loadapikeys/`.
  - 1. Rollback: remove the two `main()` calls (+ any `loader.go` precedence change); loader returns to DEAD.
- **T1.1.2 — Multi-alias key-recognition table.**
  - 1. RED: `keyrecognition_test.go::TestPresentProviders_AliasAndPlacehol` — set `CLAUDE_API_KEY` (alias), assert anthropic present; set a placeholder value, assert NOT present. RED FAILS (no table exists).
  - 1. IMPL: create `internal/llm/keyrecognition.go` with `PresentProviders() map[string]bool`, the `{provider→[]envVarAlias}` map (lifted decoupled from helix_agent `provider_types.go:349`), `isPlaceholder()`.
  - 1. GREEN: unit table-driven pass; paired §1.1 mutation (drop an alias → that alias no longer recognized → test FAILs) captured.
  - 1. Rollback: delete `keyrecognition.go` + test.

Phase P1.2 — Key-presence → provider-availability gate + factory enumerator

- **T1.2.1 — Key-presence gate in the verifier path.**
  - 1. RED: `adapter_test.go::TestGetWorkingModels_KeyPresenceGate` — verifier returns OpenAI+Anthropic verified models, `present={anthropic:true}`; assert OpenAI models DROPPED. RED FAILS (no `GetWorkingModels`).
  - 1. IMPL: add `GetWorkingModels(ctx, present)` to `adapter.go`; it calls `GetVerifiedModels` then filters by `present[m.Provider]` (+ the D-4 predicate, T1.3.1).
  - 1. GREEN: predicate truth-table pass; **paired §1.1 mutation** (strip the `present[...]` check → no-key provider leaks → FAIL) captured.
  - 1. Rollback: remove `GetWorkingModels`; callers fall back to `GetVerifiedModels`.
- **T1.2.2 — Key-present provider enumerator (`factory.go:9`).**
  - 1. RED: `factory_test.go::TestBuildPresentProviders_EnumeratesAllKeyed` — present={openai,anthropic}; assert TWO providers built (current

`NewProvider` builds one by `config.Type` → RED FAILS for the set case).

  - 1. IMPL: add `BuildPresentProviders(configs, present) ([]Provider, error)` looping every key-present provider, reusing `NewProvider` per type.
  - 1. GREEN: capture the enumerated provider list; mutation (force single-provider → set incomplete → FAIL).
  - 1. Rollback: remove enumerator; `InitializeModelManager` unchanged.

Phase P1.3 — Dynamic working-model exposure (D-4, D-2, D-5)

- **T1.3.1 — Apply the working-model filter (D-4).**
  - 1. RED: `adapter_test.go::TestGetWorkingModels_VerifiedAndScore` — feed a `failed` model, a `verified` model with score 4.0 (<min 6.0), and a `verified` score 8.0; assert only the last survives. RED FAILS today (no path applies `GetMinAcceptableScore:175` / Verified / status).
  - 1. IMPL: inside `GetWorkingModels`, apply `m.Verified && m.VerificationStatus=="verified" && m.OverallScore >= a.GetMinAcceptableScore()`. Honour CONST-038 poll ≤60s + CONST-037 verified ≤24h via existing cache TTL.
  - 1. GREEN: truth-table + paired §1.1 mutation (drop the score check → low-score model leaks → FAIL).
  - 1. Rollback: filter reverts to key-presence-only.
- **T1.3.2 — CLI/server/registry exposure switch (D-2).**
  - 1. RED: `e2e/.../sp1_working_models` — set ONLY `ANTHROPIC_API_KEY`, run `cli --list-models`; assert NO `failed`/`pending` rows and NO non-Anthropic providers. RED FAILS today (`printVerifiedModels:1392` shows failed/pending; P1 lists whole catalog).
  - 1. IMPL: `handleListModels:1361` → `GetWorkingModels(ctx, present)`; `printVerifiedModels:1392` renders only working rows; server `handlers.go:1049/1140/1206` + `model_manager.go:132/202` likewise.
  - 1. GREEN: captured CLI transcript shows only Anthropic verified models; then a REAL `generate` returns REAL output (§11.4.107 liveness, not metadata). Evidence `docs/qa/<run-id>/working-models/`.
  - 1. Rollback: revert call-sites to `GetVerifiedModels` + old printer.
- **T1.3.3 — Safe verifier-disabled / cold behaviour (DECISION-2).**
  - 1. RED: `TestListModels_VerifierDisabled_NoBluff` — `HELIX_VERIFIER_ENABLED=false`; assert the command does NOT present `FallbackModels` as "working". RED FAILS today (`main.go:1387` prints fallback list).
  - 1. IMPL: per DECISION-2 — empty + honest notice (recommended) or `--include-unverified` live path.
  - 1. GREEN: capture the honest-notice output; mutation (re-enable fallback-as-working → FAIL).
  - 1. Rollback: restore P3 fallback (NOT recommended — keeps the bluff).
- **T1.3.4 — De-hardcode provider model lists (D-5 / CONST-036, DECISION-3).**
  - 1. RED: `TestProviderModels_NotHardcoded_OpenAI/Anthropic` — assert `GetModels()` does not return the exact literal slice when a live source is reachable (or that the user-facing funnel never surfaces the literal). RED FAILS today (literal slices at `openai_provider.go:201`, `anthropic_provider.go:203`).

- 1. IMPL: per DECISION-3 — funnel-first (route all listing through verifier) then convert OpenAI/Anthropic/DeepSeek/Mistral `initializeModels` to live-fetch (mirror OpenRouter `fetchCatalog` / Ollama `/api/tags`). §11.4.124: convert, don't silently delete.
- 1. GREEN: capture a live model list from a real provider endpoint; anti-bluff smoke (`grep` for `simulated|for now|placeholder`) clean.
- 1. Rollback: restore literal slices (CONST-036 violation returns).

Phase P1.4 — Docs & guides (CONST-066 / §11.4.65)

- **T1.4.1** Write `docs/guides/model-access.md` (key setup, `.env/api_keys.sh` auto-load, alias table, working-model definition, verifier-disabled behaviour); export `.html`+`.pdf`.
- **T1.4.2** Update `.env.example` (auto-load note + aliases + verifier behaviour) and `docs/ARCHITECTURE.md` (funnel flow). Update README Documentation Map (CONST-062).

Phase P1.5 — Full test-type coverage + regression guards (§11.4.135, CONST-050)

- **T1.5.1** Stress/chaos (`tests/stresschaos/sp1_model_access_stress_test.go`): verifier-unreachable → honest degradation (not fallback-as-working); 10 concurrent `GetWorkingModels`; §11.4.85 evidence.
- **T1.5.2** Standing regression guards (`tests/regression/sp1_working_model_guard_test.go`) registering D-2, D-3, D-4 with `RED_MODE` polarity per §11.4.115; wire into the release-gate suite (§11.4.40).
- **T1.5.3** Challenge bank (`challenges/banks/sp1_model_access/`) + HelixQA registration; captured bidirectional transcript under `docs/qa/<run-id>/` (§11.4.83).
- **T1.5.4** Code-review gate (§11.4.125/§11.4.134) before pre-build + main build; iterate-until-GO.

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7. Test-type matrix for SP1

Test type	Harness	Anti-bluff requirement
Unit	`keyrecognition_test.go`, `adapter_test.go`, `factory_test.go`	mocks OK (unit only); paired §1.1 mutation per filter.
Integration (`-tags=integration`)	`tests/integration/sp1_model_access_test.go`	REAL verifier (`make test-verifier-integration`) + REAL provider key; NO mocks (CONST-050).
E2E	`tests/e2e/challenges/sp1_working_models`	one-key→only-that-provider→REAL

Test type	Harness	Anti-bluff requirement
		generate returns REAL output (§11.4.107 liveness).
Full-automation	self-driving, `-count=3` re-runnable	no operator typing (§11.4.98).
Stress + chaos	`tests/stresschaos/sp1_model_access_stress_test.go`	verifier-down honest degradation; concurrency (§11.4.85).
Security	reuse `make test-security-full`	key never logged/committed (CONST-042); placeholder rejection.
Regression guard	`tests/regression/sp1_working_model_guard_test.go`	RED_MODE polarity; release-gate blocker (§11.4.135).
Challenge / HelixQA	`challenges/banks/sp1_model_access/` + HelixQA bank	captured transcript (§11.4.83).
Performance/benchmark	`go test -bench` on `GetWorkingModels`	filter is O(n), no regression.

UI/UX/DDoS/scaling: N/A-or-delegated for this server/CLI-only surface — record honest coverage note (§11.4.3), do not fake (SP7 owns the cross-cutting matrix).

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8. Docs & guides deliverables

- `docs/guides/model-access.md` (+`.html`/`.pdf`) — user manual.
- `.env.example` — auto-load + aliases + verifier-disabled behaviour.
- `docs/ARCHITECTURE.md` — funnel flow.
- README Documentation Map + revision header (CONST-062/064).
- `docs/Issues.md`/`Issues_Summary.md`/`Fixed.md`/`CONTINUATION.md` — ATM-NNN entries, type-aware closure (CONST-057), kept in sync each state-advancing commit (CONST-044/063).

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9. Reuse/lift ledger

Per §11.4.74 (record in tracker `Catalogue-Check:`): - **Key-recognition table** → `extend helix_agent@<sha>` — lift the decoupled `{provider→[]envVar}` map from `provider_types.go:349 SupportedProviders.EnvVars` into `internal/llm/keyrecognition.go` (DECISION-4(b)); converge on the shared substrate when D-7 (`replace dev.helix.agent`) lands under SP4/SP5. - **Key→discover flow** → `reuse helix_agent@<sha>` pattern from `startup.go:389` (per-env-var → DiscoverModels dynamic-first) as the de-hardcoding template (T1.3.4). - **Faulty-key registry / scanner** → `reuse llms_verifier/api_keys@<sha>` (`env_scanner.go/manager.go/priority.go`) when D-7 wiring exists; until then, key-presence is sufficient for SP1. - **Verifier client/adapter** → already real (`verifier/client.go` live HTTP) — extend, don't replace.

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10. Summary

1. SP1 makes a provider's WORKING models appear iff a key is recognized (`.env/api_keys.sh`/shell) AND the verifier validates them — closing D-2, D-3, D-4, D-5 plus the no-key-gate and single-factory gaps.
2. Root cause is wiring, not new architecture: `secrets.LoadAPIKeys` (loader.go:30) is DEAD; the working-model predicate exists in pieces (`GetMinAcceptableScore:175`, `VerifiedModel` fields) but is never assembled.
3. The fix adds a `keyrecognition.go` present-providers table (lifted decoupled from helix_agent `SupportedProviders.EnvVars:349`), a `GetWorkingModels` filter (key∧Verified∧status∧min-score), a key-present factory enumerator, and routes CLI/server/registry through the funnel.
4. De-hardcoding (CONST-036) converts OpenAI/Anthropic/DeepSeek/Mistral literal lists to live/verifier-sourced; §11.4.124 satisfied — convert, never silently delete (CLI P2 reaches them, so not dead code).
5. Verifier-disabled MUST NOT present `FallbackModels` as "working" (that is a CONST-035/§11.4 PASS-bluff) — DECISION-2 specifies honest empty-or-unverified behaviour.
6. Every task is TDD RED-first (§11.4.43/§11.4.115) with a `RED_MODE` polarity switch and a paired §1.1 mutation; every closure registers a standing regression guard (§11.4.135).
7. Tests beyond unit hit REAL verifier + REAL provider endpoints (CONST-050); "working" models are proven by a REAL generate returning REAL output (§11.4.107 liveness), never metadata-only.
8. Four operator decisions are flagged (precedence, verifier-disabled behaviour, de-hardcode strategy, table source) — surfaced via AskUserQuestion (§11.4.66) BEFORE the affected tasks; no guessing (§11.4.6).
9. Deliverables include docs/guides (`model-access.md`, `.env.example`, ARCHITECTURE), the full SP1 test-type matrix, Challenge/HelixQA banks with captured transcripts, and synced Issues/Continuation (CONST-044/063/066).
10. Reuse-first per §11.4.74: extend helix_agent + llms_verifier rather than re-implement; converge on the shared substrate when the SP4/SP5 `replace dev.helix.agent` wiring (D-7) lands.

Ordered task count: 18 tasks across 6 phases (P1.0: 3, P1.1: 2, P1.2: 2, P1.3: 4, P1.4: 2, P1.5: 4) + 4 operator decisions (DECISION-1..4).